

NETWORK SECURITY

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Network security protects networks and the data they carry from unauthorized access, misuse, and cyberattacks. It ensures systems remain confidential, available, and trustworthy across all digital environments. Its features are:

- Prevents unauthorized access and cyber threats
- Protects data integrity, confidentiality, and availability
- Uses layered security controls across devices, users, and systems
- Ensures safe communication and reliable network operations

FIREWALL BASICS:

A Firewall is a network security system, available as hardware or software, that monitors and controls incoming and outgoing traffic based on predefined rules. It acts like a security guard, filtering data packets to either:

- Accept: Allow the traffic
- Reject: Block with an error response
- Drop: Block silently without response

Types of Firewall Based on Traffic Filtering Method

1. Packet Filtering Firewall: A basic firewall that checks packet headers like IP, port, and protocol.

- Very fast and lightweight
- Does not inspect data inside packets
- Provides basic security only

2. Stateful Inspection Firewall: Tracks active connections and makes decisions based on traffic context.

- More secure than packet filtering
- Remembers past traffic (state table)
- Blocks suspicious or unexpected packets

3. Proxy (Application-Level) Firewall: Acts as a middleman between user and destination server.

- Filters data at the application layer
- Hides internal network details
- Can block malicious content before it reaches the user

4. Circuit-Level Gateway: Validates session creation (like TCP handshake) without inspecting data.

- Ensures only legitimate sessions start
- Low processing overhead
- Does not detect internal packet threats

5. Next-Generation Firewall (NGFW): Advanced firewall combining multiple security features.

- Includes intrusion prevention, malware detection
- Can inspect encrypted traffic
- Provides app-level filtering

Basic IP Tables Rules On Linux (Allow/Deny Ports, Allow SSH)

iptables is the Linux kernel's built-in packet filtering firewall

Creating a Defensive Perimeter — "Default deny": `iptables -P INPUT DROP`

Stateful Filtering — "Only allow responses, not new connections": `iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT`

Pentest-Specific — Maintaining Your C2 Access: `iptables -A INPUT -p tcp --dport 4444 -m conntrack --ctstate NEW -j ACCEPT`

Logging for Reconnaissance — "See who's knocking": `iptables -A INPUT -j LOG --log-prefix "IPT-DROP: "`

Rate Limiting — Thwarting Brute Force: `iptables -A INPUT -p tcp --dport 22 -m recent --update --seconds 60 --hitcount 4 -j DROP`

```
ubuntu@ubuntu-virtual-machine:~$ sudo su
[sudo] password for ubuntu:
root@ubuntu-virtual-machine:/home/ubuntu# iptables -P INPUT DROP
root@ubuntu-virtual-machine:/home/ubuntu# iptables -A INPUT -m conntrack --ctstate ESTABLISHED,RELATED -j ACCEPT
root@ubuntu-virtual-machine:/home/ubuntu# # On YOUR listener machine - allow incoming on your listener ports
iptables -A INPUT -p tcp --dport 4444 -m conntrack --ctstate NEW -j ACCEPT
root@ubuntu-virtual-machine:/home/ubuntu# iptables -A INPUT -j LOG --log-prefix "IPT-DROP: "
root@ubuntu-virtual-machine:/home/ubuntu# iptables -A INPUT -p tcp --dport 22 -m recent --update --seconds 60 --hitcount 4 -j DROP
root@ubuntu-virtual-machine:/home/ubuntu#
```

All these commands run successfully.

Verification:

1. View all current rules using `sudo iptables -L -n -v`

```
root@ubuntu-virtual-machine:/home/ubuntu#
sudo iptables -L -n -v
Chain INPUT (policy DROP 285 packets, 20044 bytes)
 pkts bytes target    prot opt in     out     source         destination
    7   733 ACCEPT    all  --  *      *       0.0.0.0/0      0.0.0.0/0
        ctstate RELATED,ESTABLISHED
    0     0 ACCEPT    tcp  --  *      *       0.0.0.0/0      0.0.0.0/0
        tcp dpt:4444 ctstate NEW
   195 11286 LOG      all  --  *      *       0.0.0.0/0      0.0.0.0/0
        LOG flags 0 level 4 prefix "IPT-DROP: "
    0     0 DROP     tcp  --  *      *       0.0.0.0/0      0.0.0.0/0
        tcp dpt:22 recent: UPDATE seconds: 60 hit_count: 4 name: DEFAULT side:
source mask: 255.255.255.255

Chain FORWARD (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination

Chain OUTPUT (policy ACCEPT 0 packets, 0 bytes)
 pkts bytes target    prot opt in     out     source         destination
```

2. View a specific chain `sudo iptables -L INPUT -n -v`

```
root@ubuntu-virtual-machine:/home/ubuntu# sudo iptables -L INPUT -n -v
Chain INPUT (policy DROP 345 packets, 23620 bytes)
 pkts bytes target    prot opt in     out     source         destination
   14  1824 ACCEPT    all  --  *      *       0.0.0.0/0      0.0.0.0/0
        ctstate RELATED,ESTABLISHED
    0     0 ACCEPT    tcp  --  *      *       0.0.0.0/0      0.0.0.0/0
        tcp dpt:4444 ctstate NEW
   255 14862 LOG      all  --  *      *       0.0.0.0/0      0.0.0.0/0
        LOG flags 0 level 4 prefix "IPT-DROP: "
    0     0 DROP     tcp  --  *      *       0.0.0.0/0      0.0.0.0/0
        tcp dpt:22 recent: UPDATE seconds: 60 hit_count: 4 name: DEFAULT side:
source mask: 255.255.255.255
```

The pkts column shows how many packets have matched each rule so far.

IDS and IPS systems:

Intrusion Detection Systems (IDS) and Intrusion Prevention Systems (IPS) are two critical network security tools used to protect systems and networks from cyber threats. While they both aim to safeguard against cyberattacks, they differ in how they operate and the actions they take.

Intrusion Detection Systems (IDS)

IDS is a hardware or software tool that monitors and analyzes network or system activities for signs of unauthorized access or policy violations. It passively scans incoming traffic and compares it to pre-

configured signatures or behavioral patterns. IDS alerts the system administrator when potential threats are detected but does not take any active action to block or mitigate them.

Intrusion Prevention Systems (IPS)

IPS is an advanced security system that not only detects malicious activities but also prevents them from happening in real-time. It intercepts network traffic, compares it to known threat signatures, and immediately blocks or neutralizes suspicious activities before they can cause damage.

Tools for IDS and IPS:

- **Suricata** is a free, open-source IDS/IPS cybersecurity tool that acts as both an Intrusion Detection System (IDS) and an Intrusion Prevention System (IPS). Suricata's strength lies in its versatility. When tuned correctly, it is a high-performance tool that can handle large volumes of network traffic and generate vast amounts of network traffic data. It is also extremely flexible, offering deep analysis of various protocols and the ability to customize rule sets.
- **Snort** is a widely-used, free, and open-source intrusion detection tool for network security. It can also be configured to act as an Intrusion Prevention System (IPS). Snort monitors network traffic, analyzes packets to search for malicious content using a rule-based system to identify potential threats, and alerts or blocks traffic based on its findings.

VPN

A Virtual Private Network (VPN) is a security technology that creates an encrypted tunnel between your device and a VPN server over the internet, so your traffic travels privately and your real IP address is hidden.

- **Privacy protection:** Hides your IP and encrypts traffic, so ISPs, advertisers, and third parties can't easily monitor your browsing.
- **Security on public Wi-Fi:** Encryption protects logins and personal data on unsafe networks (airports, cafes).
- **Bypass geo-restrictions:** Changes your apparent location by routing through another region's server, unlocking region-limited sites or services.
- **Reduce ISP throttling:** Since traffic is encrypted, ISPs can't easily identify specific activities (streaming/gaming) to selectively slow them.
- **Secure remote access:** Let employees connect to internal company resources securely from outside the organization.

How VPN works to protect network traffic:

Connection establishment: When you turn on the VPN, the VPN client authenticates you and sets up a secure session with a VPN server.

Data encryption: Your outgoing traffic is encrypted using cryptographic algorithms, so intercepted packets look like unreadable cipher text to hackers, ISPs, or attackers on public Wi-Fi.

Traffic redirection (tunneling): The encrypted traffic is sent through the tunnel to the VPN server, and the VPN server replaces your real IP with its own public IP, hiding your identity/location from the destination site.

Decryption and forwarding: The VPN server decrypts the traffic and forwards it to the actual website/service; replies come back to the VPN server and then travel back to you through the same encrypted tunnel.

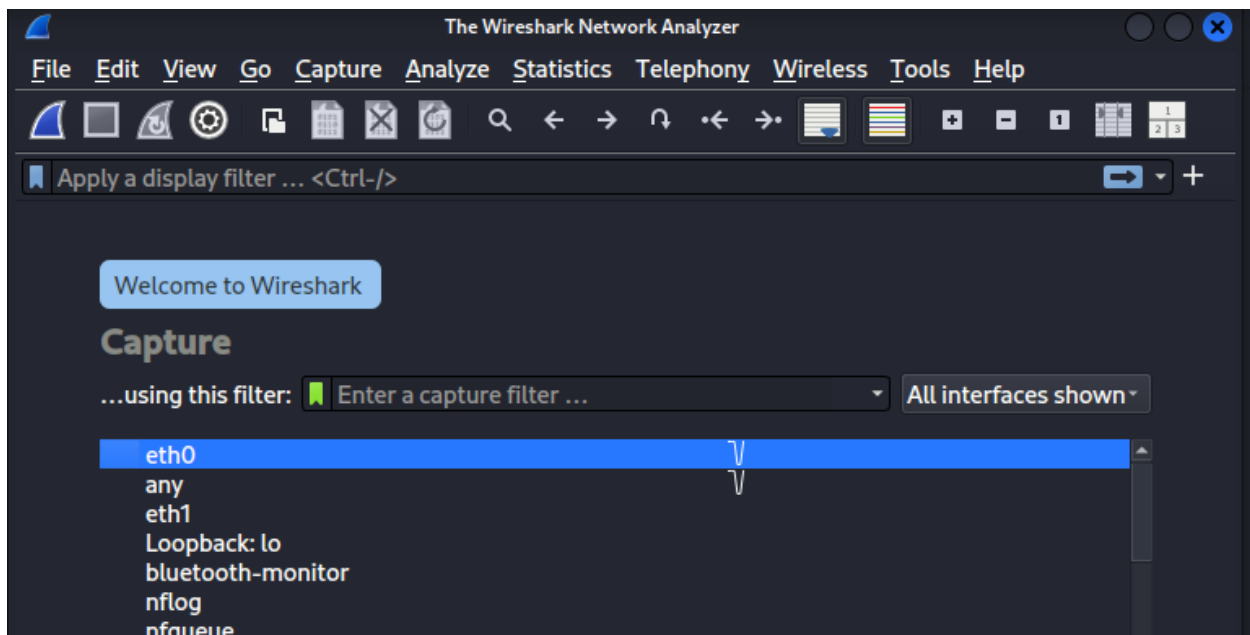
End-to-end protection: This tunnel provides privacy (confidentiality), helps maintain integrity against tampering during transit, and improves anonymity by masking your IP.

Wireshark Packet Analysis On Sample Pcap File

Opening wireshark

```
(root@kali123)-[/home/kali123]  
# sudo wireshark
```

Select eth0 and press capture



Generate traffic by these commands

```
(root@kali123)-[/home/kali123]
# # In a terminal, while Wireshark is running:
ping google.com # ICMP packets
curl http://example.com # HTTP request
ssh user@target # SSH connection
nslookup google.com # DNS query
nmap -sS 192.168.1.1 # SYN scan (generates lots of packets)
nc -lvp 4444 # Start a listener

PING google.com (142.250.200.174) 56(84) bytes of data.
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=1 ttl=128 time=43.4 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=2 ttl=128 time=43.1 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=3 ttl=128 time=44.1 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=4 ttl=128 time=44.0 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=5 ttl=128 time=43.6 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=6 ttl=128 time=43.6 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=7 ttl=128 time=43.4 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=8 ttl=128 time=47.7 ms
64 bytes from pnfjra-aq-in-f14.1e100.net (142.250.200.174): icmp_seq=9 ttl=128 time=43.1 ms
```

Traffic generated and saved as .pcapng file

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.190.130	142.251.37.142	ICMP	98	Echo (ping) request id=0x0003, seq=41/10496, t
2	0.053476801	142.251.37.142	192.168.190.130	ICMP	98	Echo (ping) reply id=0x0003, seq=41/10496, t
3	0.220409412	VMware_46:df:e5	Broadcast	ARP	42	Who has 192.168.190.133? Tell 192.168.190.130
4	0.321298689	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
5	1.002621591	192.168.190.130	142.251.37.142	ICMP	98	Echo (ping) request id=0x0003, seq=42/10752, t
6	1.068383735	142.251.37.142	192.168.190.130	ICMP	98	Echo (ping) reply id=0x0003, seq=42/10752, t
7	1.130367289	VMware_46:df:e5	VMware_ea:eb:c5	ARP	42	Who has 192.168.190.2? Tell 192.168.190.130
8	1.130873008	VMware_ea:eb:c5	VMware_46:df:e5	ARP	60	192.168.190.2 is at 00:50:56:ea:eb:c5
9	1.226613678	VMware_46:df:e5	Broadcast	ARP	42	Who has 192.168.190.133? Tell 192.168.190.130
10	1.306803853	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
11	2.005534225	192.168.190.130	142.251.37.142	ICMP	98	Echo (ping) request id=0x0003, seq=43/11008, t

Identifying Suspicious Traffic Patterns

Apply filters:

1. For arp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
2	1.155276341	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
3	2.006923614	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
4	3.015309883	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
5	4.195398714	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
6	4.997236370	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
7	5.999909695	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
8	7.228302204	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
9	8.003913232	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
10	9.000689415	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1
11	10.250489707	VMware_c0:00:08	Broadcast	ARP	60	Who has 192.168.190.2? Tell 192.168.190.1

2. Used dns filter

dns						
No.	Time	Source	Destination	Protocol	Length	Info
770	305.743186183	192.168.190.130	192.168.190.2	DNS	88	Standard query 0x6f7b A contile.services.mozill
771	305.743545196	192.168.190.130	192.168.190.2	DNS	88	Standard query 0xf378 AAAA contile.services.moz
774	305.953665709	192.168.190.130	192.168.190.2	DNS	79	Standard query 0xc1d2 A spocs.getpocket.com
775	305.953789577	192.168.190.130	192.168.190.2	DNS	79	Standard query 0x93d0 AAAA spocs.getpocket.com
776	306.648962121	192.168.190.130	192.168.190.2	DNS	95	Standard query 0x7889 A content-signature-2.cdn
777	306.649052999	192.168.190.130	192.168.190.2	DNS	95	Standard query 0x138c AAAA content-signature-2.c
778	306.698851142	192.168.190.2	192.168.190.130	DNS	111	Standard query response 0x7889 A content-signatu
779	306.698851672	192.168.190.2	192.168.190.130	DNS	123	Standard query response 0x138c AAAA content-sig
798	307.283688763	192.168.190.130	192.168.190.2	DNS	87	Standard query 0x28a8 A safebrowsing.googleapis
799	307.283814243	192.168.190.130	192.168.190.2	DNS	87	Standard query 0x00a6 AAAA safebrowsing.googleap
800	307.295223176	192.168.190.2	192.168.190.130	DNS	103	Standard query response 0x28a8 A safebrowsing.g
800	307.295789910	192.168.190.2	192.168.190.130	DNS	115	Standard query response 0x00a6 AAAA safebrowsin
▶ Frame 770: 88 bytes on wire (704 bits), 88 bytes captured (704 bits) on interface 0 ▶ Ethernet II, Src: VMware_46:df:e5 (00:0c:29:46:df:e5), Dst: Vmware_46:df:e5 (00:0c:29:46:df:e5) ▶ Internet Protocol Version 4, Src: 192.168.190.130, Dst: 192.168.190.2 ▶ User Datagram Protocol, Src Port: 52392, Dst Port: 53 ▶ Domain Name System (query)						
				0000	00 50 56	ea eb c5 00 0c 29 46 df e5 08 00 45 00 .PV..
				0010	00 4a 36 77 40 00 40 11	06 56 c0 a8 be 82 c0 a8 .J6w@
				0020	be 02 cc a8 00 35 00 36	fe 1d 6f 7b 01 00 00 01
				0030	00 00 00 00 00 00 07 63	6f 6e 74 69 6c 65 08 73
				0040	65 72 76 69 63 65 73 07	6d 6f 7a 69 6c 6c 61 03
				0050	63 6f 6d 00 00 01 00 01	com...